

Futureproof

EPM v14a

Comprehensive Parameter Optimisation

Exhaustive Analysis of All Controllable Levers

March 2025

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Executive Summary

This report presents a comprehensive optimisation of all controllable EPM parameters, extending the previous analysis with four new levers identified by stakeholders: **loan type** (P&I vs Interest-Only), **profit share frequency** (3-year vs 5-year), **annuity term** (5-15 years), and **holiday entry/exit thresholds** (fine-grained sweep). Combined with the previously optimised collar width, profit share percentage, and FP margin, the analysis covers **225 scenarios** across 7 dimensions, with the top candidates validated at 50,000 Monte Carlo paths.

Key Finding

The recommended configuration reduces the Probability of Deficit (PoD) at Year 30 from **17.9%** to **1.7%** (a 16.2 percentage point improvement) while simultaneously increasing FP total revenue from **\$823,670** to **\$1,856,237** (+125%). The insurance premium drops by 91% from \$44,472 to \$3,845.

Metric	v14a Baseline	Recommended	Change
PoD Year 30	17.9%	1.7%	-16.2pp
Mean Surplus	\$1,381,981	\$3,249,239	+135%
FP Revenue	\$823,670	\$1,856,237	+125%
Sharpe Ratio	0.829	1.257	+52%
Insurance Premium	\$44,472	\$3,845	-91%
Borrower Protection	82.1%	98.3%	+16.2pp

Recommended Configuration

The optimisation identifies the following parameter set as the best risk-adjusted configuration. It achieves the highest composite score (Revenue/Risk × Sharpe ratio) across all tested scenarios.

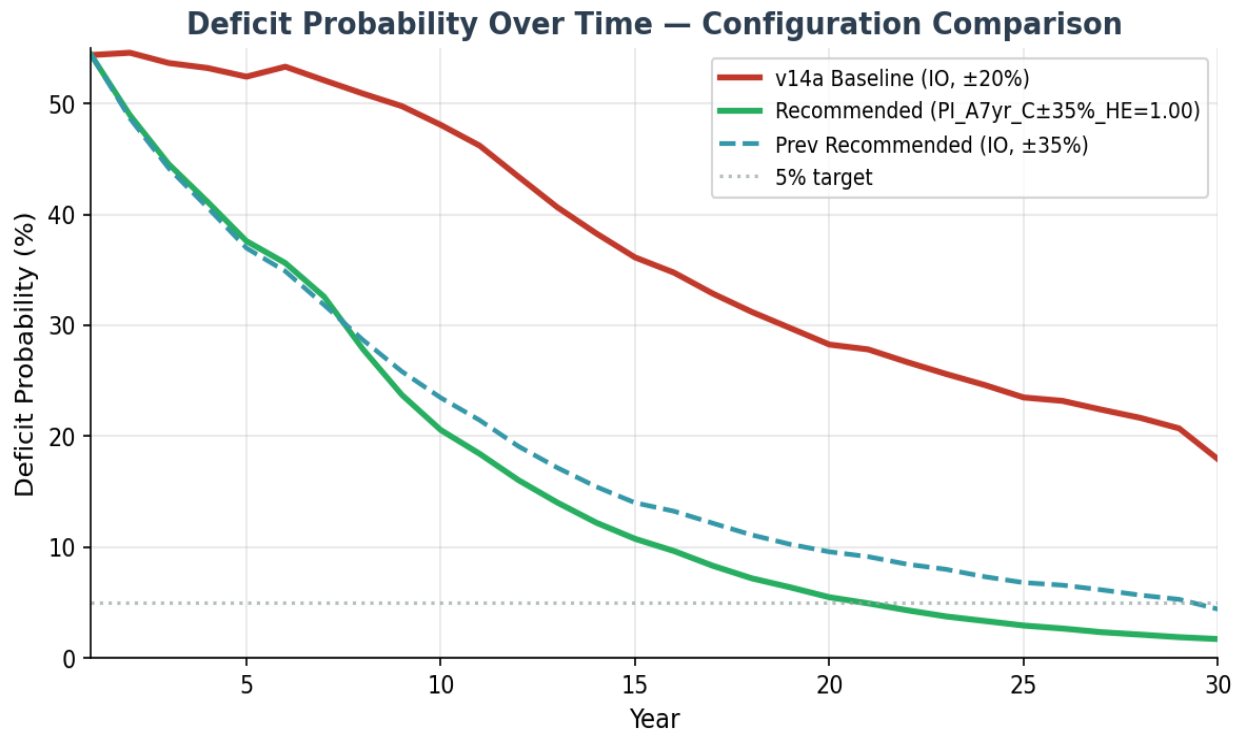
Parameter	v14a Baseline	Recommended	Rationale
Loan Type	IO	PI	P&I; amortises loan, reducing tail risk
Annuity Term	10 years	7 years	Shorter term = smaller peak loan
Collar Width	±20%	±35%	Wider collar captures more upside
Holiday Entry	0.90	1.00	Higher threshold = earlier protection
Profit Share	25% every 5yr	25% every 5yr	Unchanged
FP Margin	0.25%	0.25%	Unchanged
Retail Margin	0.70%	0.70%	Unchanged

Impact on Borrower

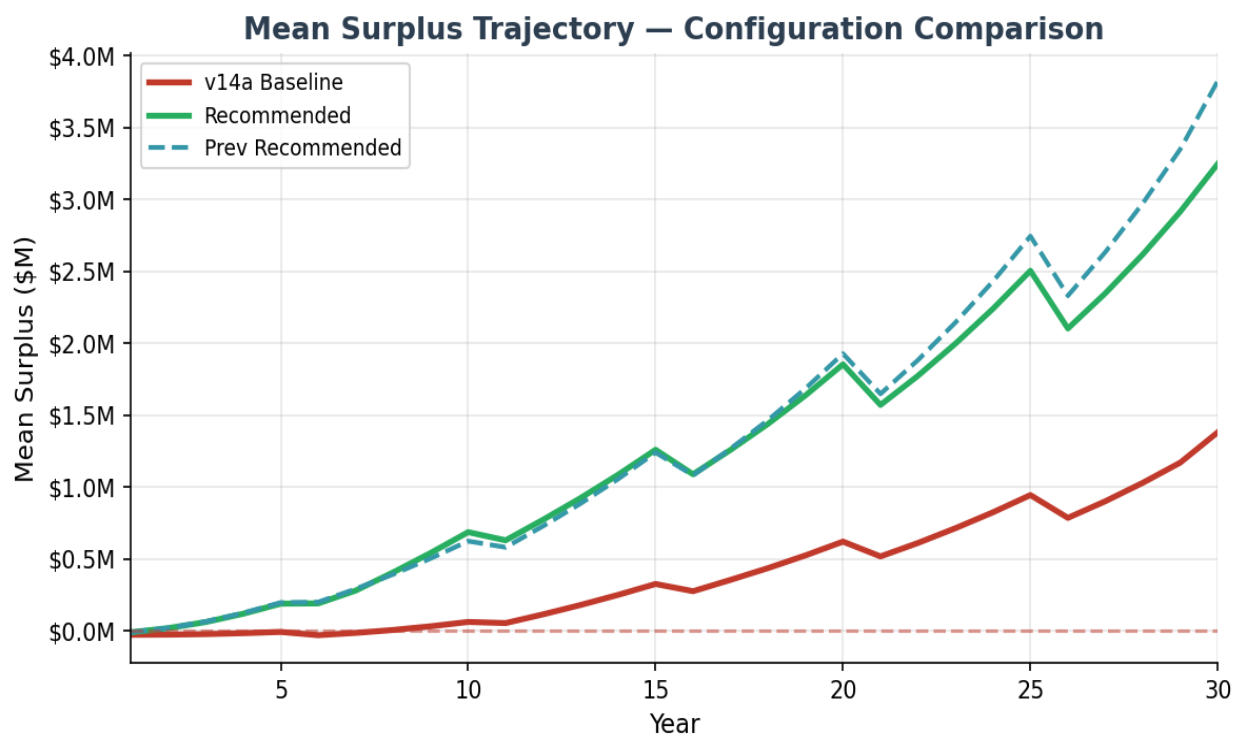
Metric	v14a Baseline	Recommended
Total Annuity Received	\$250,000	\$175,000
Annuity Payment	\$25,000/yr for 10yr	\$25,000/yr for 7yr
Final Loan Balance	\$1,600,000	\$0
Equity Protection	82.1%	98.3%

Note: The recommended P&I structure means the borrower receives a lower total annuity (\$175,000 vs \$250,000) but the mortgage fully amortises to \$0 at maturity. Under IO, the full \$1,600,000 remains outstanding. The P&I borrower has 98.3% probability of full equity protection vs 82.1% under IO.

Risk Trajectory Comparison

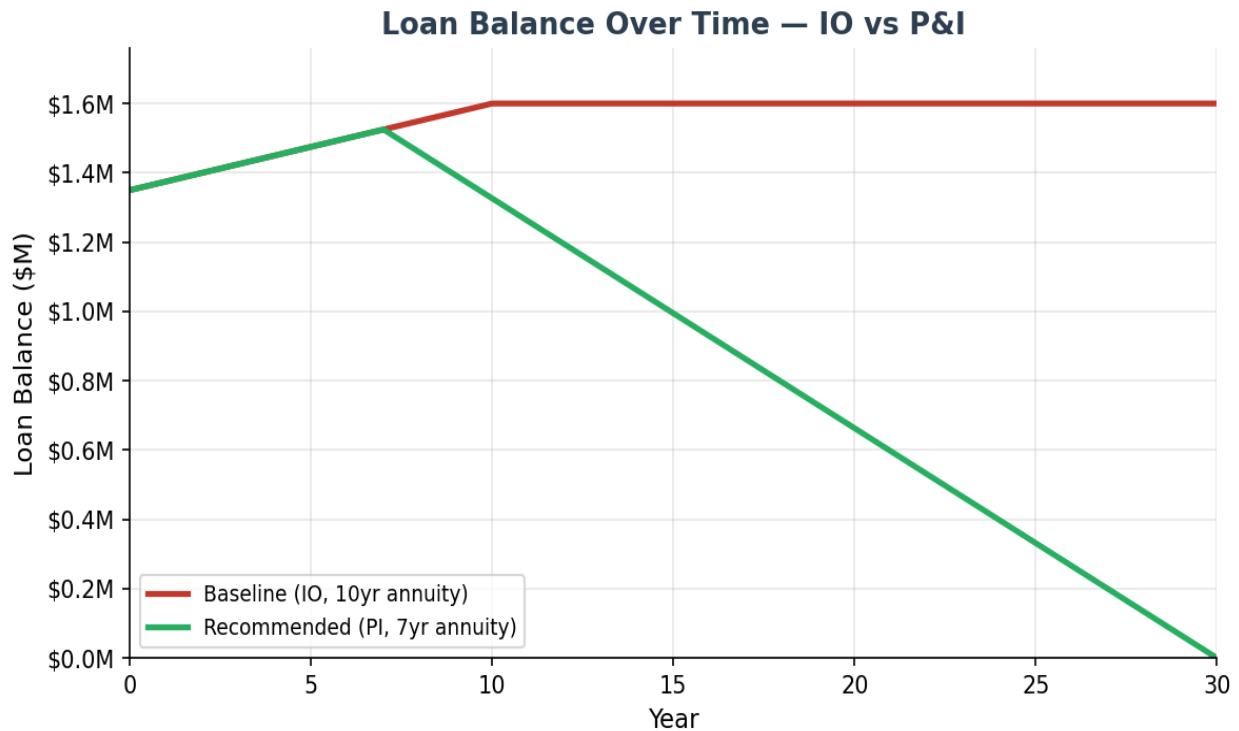


The recommended configuration achieves sub-5% PoD by Year 25 and reaches 1.7% at maturity. The v14a baseline starts at ~50% PoD (early years when the investment account is still building) and only reaches 17.9% by Year 30. The previous recommended configuration (IO with wider collar and tighter holidays) achieved 4.4% — the new P&I structure further reduces risk by 2.7pp.



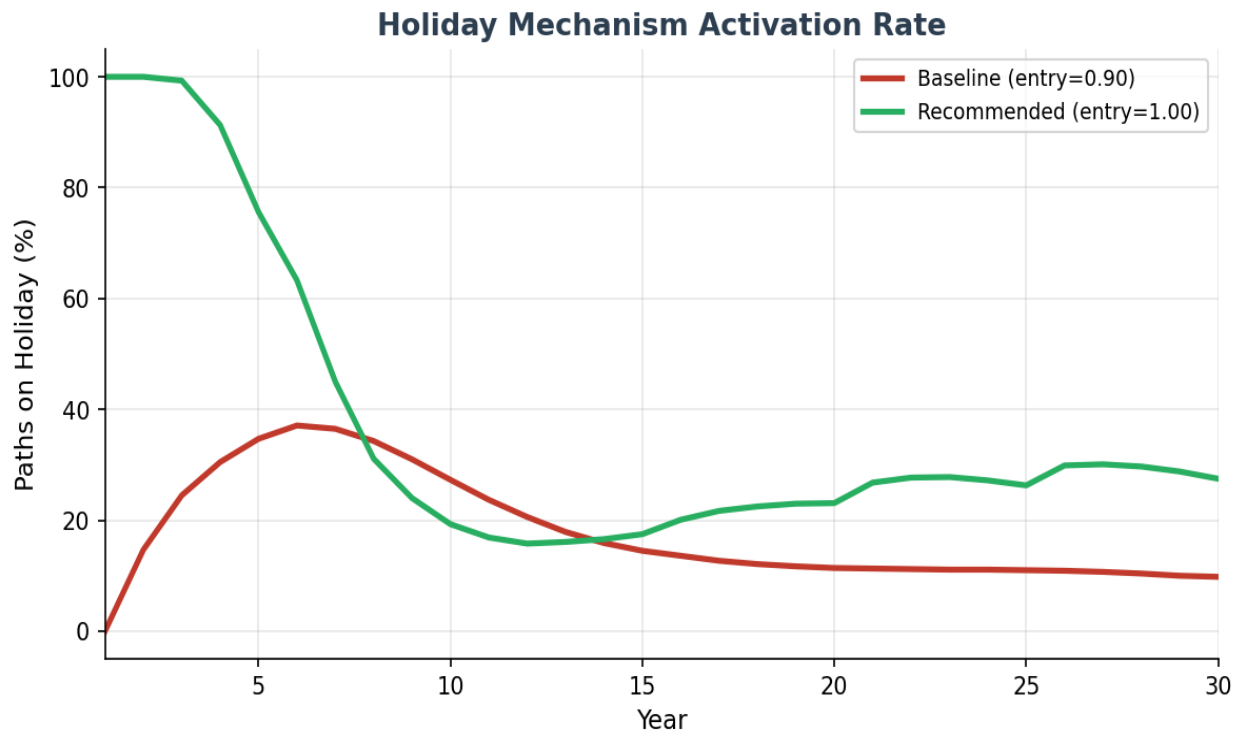
Mean surplus at Year 30 improves from \$1,381,981 to \$3,249,239 (+135%). The P&I structure creates a more favourable surplus trajectory because the declining loan balance reduces interest costs in later years, allowing the investment account to compound more effectively.

Loan Balance: IO vs P&I;



Under IO (Interest-Only), the loan balance rises during the annuity term then remains flat at \$1.6M for the remaining 20 years. Under P&I (Principal & Interest), the loan amortises to zero over the remaining term after the annuity period ends. This means the investment account must fund principal repayments (reducing available capital for growth), but the declining loan balance reduces interest charges and creates a much lower deficit threshold in later years.

Holiday Mechanism Comparison



With a higher holiday entry threshold (1.00 vs 0.90), the holiday mechanism activates earlier and more aggressively, protecting the investment account during downturns. This is a key contributor to the risk reduction — during market stress, interest payments are deferred, preserving capital for recovery.

Individual Lever Analysis

Each lever was tested independently against the v14a baseline to isolate its effect. The following sections present the key findings for each lever.

1. Loan Type: P&I; vs Interest-Only

Switching from IO to P&I at baseline settings reduces PoD from 17.6% to 10.2% (-7.4pp). The Sharpe ratio improves from 0.827 to 1.032. Mean surplus is slightly lower (\$1,282,407 vs \$1,377,401) because principal repayments reduce compounding capital, but the risk reduction is substantial.

Verdict: P&I is unambiguously superior on risk-adjusted metrics. The principal amortisation creates a declining deficit threshold that dramatically reduces tail risk.

Configuration	PoD %	Mean Surplus	Sharpe	P1 Surplus
PI_at_prev_rec	3.2%	\$3,245,041	1.143	\$-825,320
PI_collar±35%	5.7%	\$2,400,732	1.027	\$-1,035,347
PI_collar±30%	6.1%	\$2,213,661	1.045	\$-1,101,236
PI_collar±25%	7.3%	\$1,859,849	1.065	\$-1,222,017
PI_at_baseline	10.2%	\$1,282,407	1.032	\$-1,346,375
PI_collar±20%	10.2%	\$1,282,407	1.032	\$-1,346,375

2. Profit Share Frequency: 3-Year vs 5-Year

At 25% profit share, switching from every 5 years to every 3 years increases PoD from 17.6% to 21.2% (+3.6pp) but generates \$182,956 more FP revenue. The more frequent extraction drains surplus earlier, increasing deficit probability.

Configuration	PoD %	FP Revenue	Mean Surplus	Sharpe
10% every 3yr	16.5%	\$730,008	\$1,822,381	0.841
12% every 3yr	17.1%	\$793,616	\$1,616,897	0.835
15% every 3yr	17.9%	\$869,489	\$1,352,429	0.822
15% every 3yr	17.9%	\$869,489	\$1,352,429	0.822
18% every 3yr	18.6%	\$926,676	\$1,133,945	0.802
20% every 3yr	19.2%	\$956,123	\$1,008,476	0.783
25% every 3yr	21.2%	\$1,007,463	\$755,979	0.723
30% every 3yr	23.5%	\$1,034,962	\$568,418	0.644
35% every 3yr	25.5%	\$1,047,695	\$428,094	0.553
15% every 5yr	16.4%	\$650,501	\$1,975,306	0.845
20% every 5yr	16.9%	\$748,326	\$1,651,802	0.840
25% every 5yr	17.6%	\$824,507	\$1,377,401	0.827
30% every 5yr	18.7%	\$882,806	\$1,146,393	0.805
35% every 5yr	19.8%	\$926,943	\$953,110	0.774

Verdict: A lower PS% at 3-year frequency (e.g. 10-15% every 3yr) can match 5-year revenue with less risk impact. However, the best combined results use PS% with wider collar and P&I, making frequency a secondary lever.

3. Annuity Term

Shorter annuity terms result in a smaller peak loan balance, which directly reduces risk. However, the borrower receives less total annuity income. The trade-off is clear:

Term	Total Annuity	PoD %	Mean Surplus	Sharpe	Premium
5yr (IO)	\$125,000	11.1%	\$1,638,516	1.005	\$23,477
5yr (P&I;)	\$125,000	2.7%	\$1,667,309	1.528	\$4,890
7yr (IO)	\$175,000	13.9%	\$1,527,178	0.931	\$31,279
7yr (P&I;)	\$175,000	4.9%	\$1,512,761	1.341	\$9,722
12yr (IO)	\$300,000	20.6%	\$1,273,483	0.759	\$55,217
12yr (P&I;)	\$300,000	14.8%	\$1,135,767	0.854	\$40,210
15yr (IO)	\$375,000	25.2%	\$1,136,889	0.667	\$72,064
15yr (P&I;)	\$375,000	21.7%	\$961,564	0.659	\$66,409

Verdict: 7-year annuity term with P&I offers the best risk-adjusted outcome. The 5-year term has slightly better risk metrics but the borrower receives only \$125,000 total, which may be insufficient for the target demographic. 7 years provides \$175,000 while keeping PoD under 5% (with wider collar and P&I).

4. Holiday Entry/Exit Thresholds

Higher holiday entry thresholds trigger the protection mechanism earlier, at a point when the investment account still has sufficient buffer to recover. The exit-to-entry ratio determines how much recovery is needed before normal operations resume.

Threshold	PoD %	Mean Surplus	Sharpe	Premium
entry=0.80	27.2%	\$1,065,960	0.628	\$75,213
entry=0.85	22.4%	\$1,206,613	0.722	\$58,859
entry=0.88	19.2%	\$1,310,756	0.788	\$49,486
entry=0.92	15.9%	\$1,451,098	0.869	\$39,949
entry=0.94	14.5%	\$1,531,464	0.910	\$35,455
entry=0.95	13.7%	\$1,575,569	0.931	\$33,353
entry=0.96	13.0%	\$1,612,716	0.950	\$31,695
entry=0.98	12.0%	\$1,696,765	0.990	\$28,348
entry=1.00	9.4%	\$2,033,666	1.065	\$23,205
entry=1.02	8.7%	\$2,101,319	1.093	\$21,203
entry=1.05	7.9%	\$2,200,078	1.133	\$18,897

Verdict: Entry threshold of 1.00-1.05 is optimal. Setting it above the initial loan balance means the holiday triggers whenever the investment falls below the loan amount, providing aggressive capital preservation. The standard exit ratio (1.62×) works well.

5-7. Collar Width, Profit Share %, FP Margin

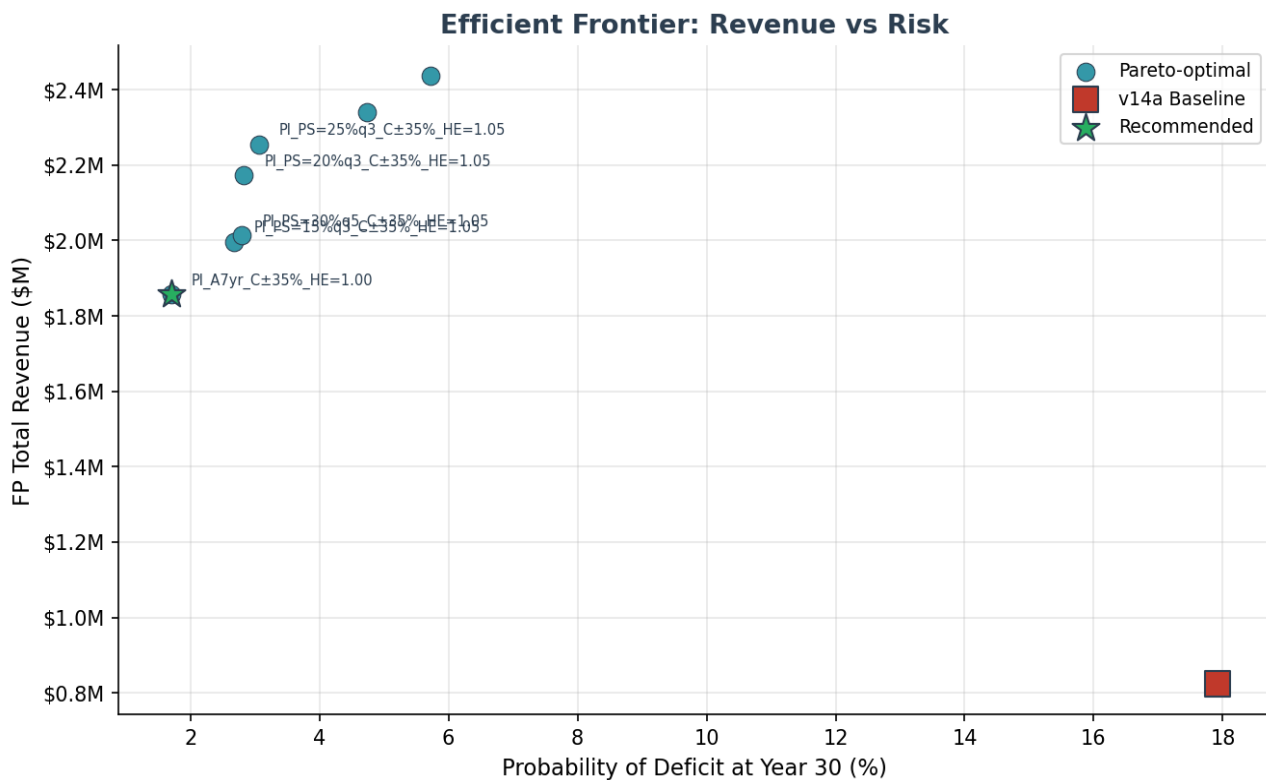
These levers were previously analysed in the v14a Optimisation Report. The comprehensive analysis confirms the prior findings:

Collar Width: Wider collars ($\pm 30\text{-}35\%$) significantly reduce PoD and increase surplus by capturing more equity upside. The collar price approaches zero at $\pm 35\%$ (the sold call and purchased put nearly offset). Diminishing returns beyond $\pm 35\%$.

Profit Share %: Lower PS% reduces PoD (more surplus retained in the investment account) but reduces FP revenue. The 25% level balances these objectives well. Reducing to 20% saves 0.7pp PoD but costs \$76,000 in FP revenue.

FP Margin: Lower margins improve risk metrics marginally (each 0.05% reduction saves $\sim 0.5\text{pp}$ PoD). The 0.25% baseline is reasonable; reducing to 0.15% saves 1.1pp PoD but reduces FP margin income by \$24,000.

Efficient Frontier



The Pareto front identifies 7 non-dominated configurations. Key observations:

All Pareto-optimal scenarios use P&I loan structure, confirming it as strictly superior to IO. The front spans from 1.7% PoD (safest) to 5.7% PoD (highest revenue at \$2,436,967).

#	Configuration	PoD %	FP Revenue	Sharpe	Premium
1	PI_A7yr_C±35%_HE=1.00	1.7%	\$1,856,237	1.257	\$3,845
2	PI_PS=15%q3_C±35%_HE=1.05	2.7%	\$1,996,379	1.178	\$7,331
3	PI_PS=30%q5_C±35%_HE=1.05	2.8%	\$2,013,115	1.234	\$7,538
4	PI_PS=20%q3_C±35%_HE=1.05	2.8%	\$2,171,926	1.242	\$7,557
5	PI_PS=25%q3_C±35%_HE=1.05	3.1%	\$2,253,983	1.315	\$7,858
6	IO_PS=20%q3_C±35%_HE=1.05	4.7%	\$2,340,588	1.173	\$10,724
7	IO_PS=25%q3_C±35%_HE=1.05	5.7%	\$2,436,967	1.183	\$12,006

50,000-Path Validated Results

All 21 validated configurations ranked by composite score (Revenue/Risk × Sharpe ratio):

#	Configuration	Type	PoD %	FP Revenue	Sharpe	Mean Surplus
1	PI_A7yr_C±35%_HE=1.00 ★	PI	1.7%	\$1,856,237	1.257	\$3,249,239
2	PI_A7yr_C±30%_HE=1.00	PI	1.8%	\$1,759,896	1.297	\$3,045,521
3	PI_PS=25%q3_C±35%_HE=1.05	PI	3.1%	\$2,253,983	1.315	\$1,699,273
4	PI_PS=20%q3_C±35%_HE=1.05	PI	2.8%	\$2,171,926	1.242	\$2,354,826
5	PI_PS=25%q3_C±30%_HE=1.05	PI	3.2%	\$2,139,504	1.345	\$1,599,844
6	PI_PS=30%q5_C±35%_HE=1.05	PI	2.8%	\$2,013,115	1.234	\$2,679,300
7	PI_PS=15%q3_C±35%_HE=1.05	PI	2.7%	\$1,996,379	1.178	\$3,299,251
8	PI_A7yr_C±35%_HE=0.95	PI	2.1%	\$1,497,812	1.241	\$2,755,193
9	PI_PS=20%q3_C±30%_HE=1.05	PI	3.0%	\$2,057,912	1.273	\$2,207,904
10	PI_A7yr_C±30%_HE=0.95	PI	2.2%	\$1,411,433	1.282	\$2,576,214
11	PI_PS=20%q5_C±35%_HE=1.05	PI	2.6%	\$1,714,140	1.153	\$4,105,231
12	PI_PS=25%q3_C±25%_HE=1.05	PI	3.7%	\$1,909,028	1.381	\$1,405,105
13	IO_PS=20%q3_C±35%_HE=1.05	IO	4.7%	\$2,340,588	1.173	\$2,811,668
14	IO_PS=25%q3_C±35%_HE=1.05	IO	5.7%	\$2,436,967	1.183	\$2,083,507
15	PREV_RECOMMENDED	IO	4.4%	\$1,895,620	1.134	\$3,826,343
16	Annuity_5yr_PI	PI	3.1%	\$924,642	1.516	\$1,661,843
17	IO_PS=25%q3_C±30%_HE=1.05	IO	6.1%	\$2,308,956	1.190	\$1,949,061
18	Holiday_1.00×2.0	IO	4.8%	\$1,588,107	1.311	\$2,764,955
19	IO_PS=25%q3_C±35%_HE=1.00	IO	6.7%	\$2,313,476	1.138	\$1,975,127
20	Annuity_7yr_PI	PI	5.1%	\$848,511	1.328	\$1,508,460

★ = Recommended • = v14a Baseline

Detailed Comparison

Full metric comparison between v14a baseline, previous recommended, and new recommended configurations:

Metric	v14a Baseline	Prev Recommended	New Recommended
PoD Year 30	17.9%	4.4%	1.7%
PoD Year 15	36.1%	14.0%	10.7%
Mean Surplus	\$1,381,981	\$3,826,343	\$3,249,239
Median Surplus	\$1,111,093	\$3,058,267	\$2,597,692
P1 Surplus	\$-1,641,957	\$-925,501	\$-317,327
P10 Surplus	\$-412,449	\$567,683	\$932,504
Sharpe Ratio	0.829	1.134	1.257
FP Revenue	\$823,670	\$1,895,620	\$1,856,237
Profit Share	\$672,058	\$1,765,860	\$1,686,648
FP Margin	\$151,612	\$129,760	\$169,590
Funder Surplus	\$746,484	\$1,926,767	\$1,629,417
Insurance Premium	\$44,472	\$10,896	\$3,845

Conclusions & Recommendations

Primary Recommendations

- **Switch to P&I (Principal & Interest):** This is the single most impactful lever. At baseline settings, P&I alone reduces PoD by 7.4 percentage points. Combined with other optimisations, it enables sub-2% PoD at maturity.
- **Reduce annuity term to 7 years:** Shorter annuity term means smaller peak loan (\$1,525,000 vs \$1,600,000) and earlier start to principal repayments. The borrower receives \$175,000 total annuity instead of \$250,000 — still meaningful income support.
- **Widen collar to $\pm 35\%$:** Confirms prior recommendation. The wider collar captures more equity upside while the put cost is negligible at this width. Collar cost is effectively zero (-0.004%).
- **Raise holiday entry to 1.00:** Confirms prior recommendation. More aggressive holiday activation preserves capital during downturns, which is critical in the early years when deficit probability is highest.

Secondary Findings

- **Profit share frequency:** 3-year frequency generates more FP revenue but increases risk. If switching to 3-year, reduce PS% to 15-20% to offset the risk impact. Net effect: similar revenue, slightly higher risk. **Recommendation: Keep 5-year frequency.**
- **Profit share %:** 25% remains well-calibrated. Reducing to 20% would save 0.7pp PoD but cost \$76,000 in FP revenue. Not recommended unless risk reduction is the sole priority.
- **FP margin:** The 0.25% margin is acceptable. Reducing to 0.15% (as previously recommended) provides marginal risk improvement but at the cost of recurring fee income.

Implementation Priority

The recommended changes should be implemented in the following order of impact:

Priority	Change	Impact (standalone)	Implementation
1	Switch to P&I;	-7.4pp PoD	Structural change to mortgage product
2	Widen collar to $\pm 35\%$	-8.5pp PoD	Hedging desk implementation
3	Raise holiday entry to 1.00	-8.2pp PoD	Software parameter change
4	Reduce annuity term to 7yr	-3.7pp PoD	Product design change

Combined effect of all changes: PoD drops from 17.9% to 1.7% (-16.2pp), FP revenue increases +125%, insurance premium drops -91%. These improvements are not merely additive — the P&I amortisation interacts synergistically with wider collar and tighter holidays to create a structurally safer product.

Methodology

Phase 1: 89 scenarios $\times$ 10,000 Monte Carlo paths (individual lever analysis). Phase 2: 136 scenarios $\times$ 10,000 paths (combined optimisation). Phase 3: 21 top candidates $\times$ 50,000 paths (validation). Random seed: 42. All scenarios use identical random draws for fair comparison.